

```
In [2]: import pandas as pd
import numpy as np
from sklearn.cluster import KMeans
from sklearn.preprocessing import MinMaxScaler
import matplotlib.pyplot as plt
```

```
In [4]: df=pd.read_csv("Untitled spreadsheet - Sheet1 (1).csv")
df.head(6)
```

```
Out[4]:
```

	Point	x	y
0	A	1	2
1	B	1	4
2	C	3	4
3	D	5	7
4	E	3	6
5	F	6	8

```
In [8]: scale=MinMaxScaler()
scale.fit(df[["x"]])
df[["x"]]=scale.transform(df[["x"]])
scale.fit(df[["y"]])
df[["y"]]=scale.transform(df[["y"]])
```

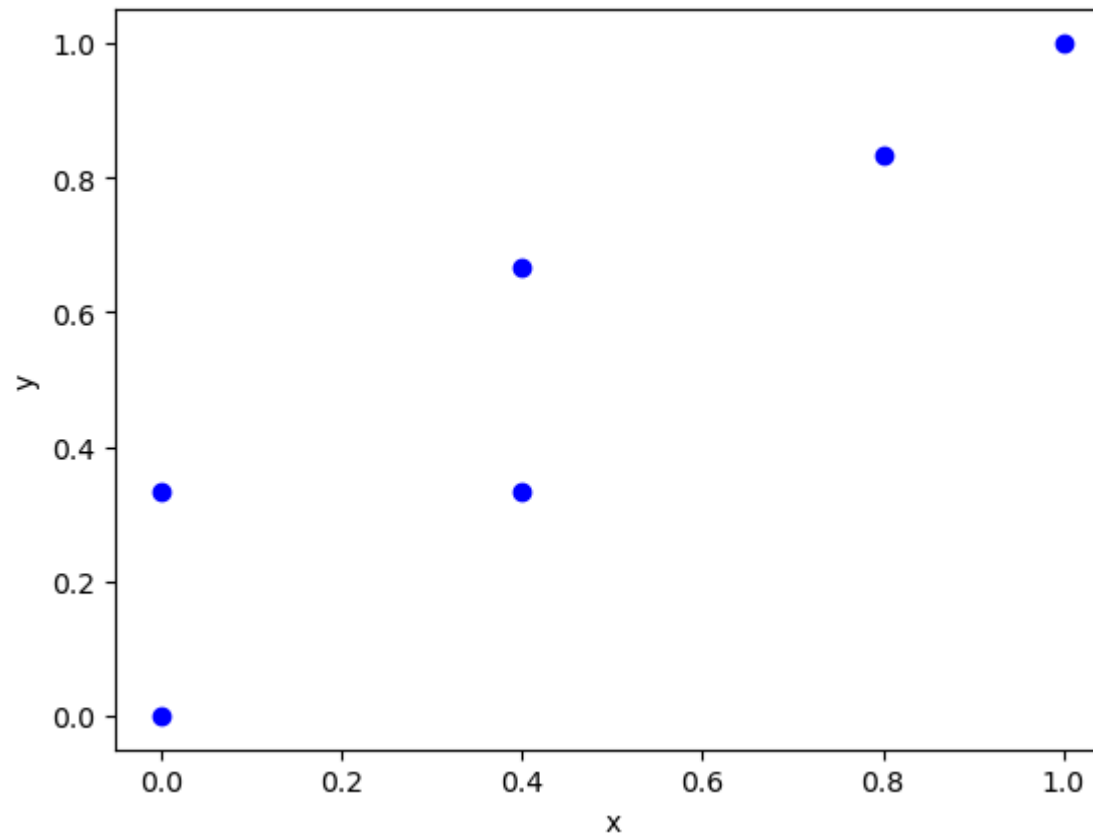
```
In [10]: df
```

Out[10]:

	Point	x	y
0	A	0.0	0.000000
1	B	0.0	0.333333
2	C	0.4	0.333333
3	D	0.8	0.833333
4	E	0.4	0.666667
5	F	1.0	1.000000

```
In [16]: plt.xlabel("x")
plt.ylabel("y")
plt.scatter(df.x,df.y,color="blue")
```

Out[16]: <matplotlib.collections.PathCollection at 0x287185c3fe0>



```
In [18]: k=KMeans(n_clusters=2)
```

```
In [20]: k.fit(df[["x", "y"]])
```

C:\Users\muham\anaconda3\Lib\site-packages\sklearn\cluster_kmeans.py:1446: UserWarning: KMeans is known to have a memory leak on Windows with MKL, when there are less chunks than available threads. You can avoid it by setting the environment variable OMP_NUM_THREADS=1.

warnings.warn(

```
Out[20]: KMeans
KMeans(n_clusters=2)
```

```
In [22]: z_pred=k.predict(df[["x","y"]])
z_pred
```

```
Out[22]: array([0, 0, 0, 1, 0, 1])
```

```
In [24]: df["clusters"]=z_pred
df
```

```
Out[24]:
```

	Point	x	y	clusters
0	A	0.0	0.000000	0
1	B	0.0	0.333333	0
2	C	0.4	0.333333	0
3	D	0.8	0.833333	1
4	E	0.4	0.666667	0
5	F	1.0	1.000000	1

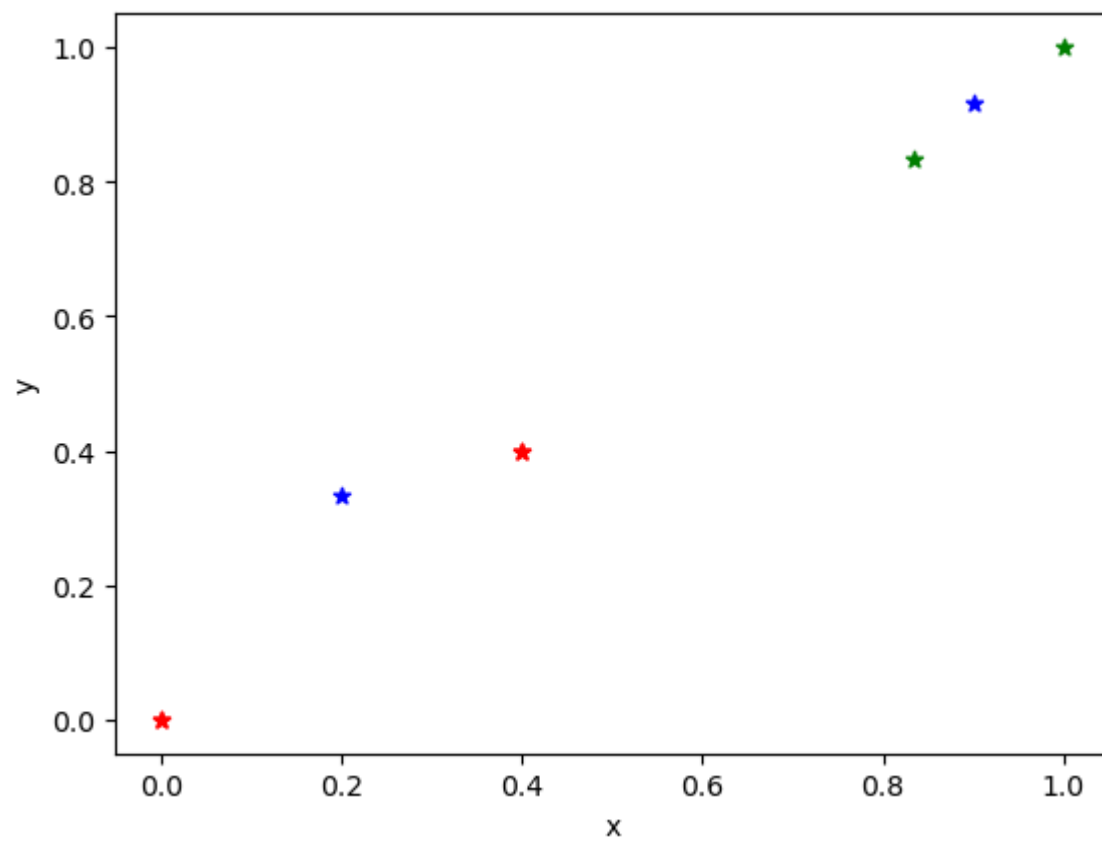
```
In [26]: data1=df[df.clusters==0]
data2=df[df.clusters==1]
```

```
In [28]: k.cluster_centers_
```

```
Out[28]: array([[0.2      , 0.33333333],
               [0.9      , 0.91666667]])
```

```
In [32]: plt.scatter(data1.x,data1.x,color="red",marker="*")
plt.scatter(data2.y,data2.y,color="green",marker="*")
plt.xlabel("x")
plt.ylabel("y")
plt.scatter(k.cluster_centers_[0],k.cluster_centers_[1],color="blue",marker="*")
```

```
Out[32]: <matplotlib.collections.PathCollection at 0x28722fe7f80>
```



In []: